

Jonathan Graehl

CURRICULUM VITAE

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Summary

Research scientist working on transformer models for machine translation, speech, and NLP: domain adaptation, word alignment, hallucination mitigation, privacy-focused named-entity recognition, inference optimization, and distillation.

Professional Experience

RWS Language Weaver (formerly SDL Research)

RESEARCH SCIENTIST

2012–present

- Transformer models for machine translation and speech translation, including domain adaptation
- Word alignment from transformer attention and encoder similarity
- Hallucination mitigation in neural and LLM-based translation
- Privacy-focused named-entity recognition model
- Inference optimization for production translation and NLP models

SDL Language Weaver

SENIOR RESEARCH ENGINEER

2011–2012

USC Information Sciences Institute

RESEARCH PROGRAMMER

2002–2011

Honors

- 2022 **ACL Test-of-Time Paper Award, for “Machine Transliteration” (ACL 1997)**
- 1998 **Computing Research Association Outstanding Undergraduate Researcher — Honorable Mention**
- 1994 **Ventura County Symphony — Discovery Artist; concerto soloist**
- 1993 **1570 SAT**

Publications

JOURNAL ARTICLES

- “The Power of Extended Top-Down Tree Transducers,” (A. Maletti, J. Graehl, M. Hopkins, and K. Knight), SIAM J. Comput., 39(2):410–430, 2009. doi
- “Training Tree Transducers,” (J. Graehl, K. Knight, and J. May), Computational Linguistics, 34(3), 2008. pdf
- “Machine Transliteration,” (K. Knight and J. Graehl), Computational Linguistics, 24(4), 1998. pdf

CONFERENCE PAPERS

- “Bayesian Inference for Finite-State Transducers,” (D. Chiang, J. Graehl, K. Knight, A. Pauls, and S. Ravi), Proc. NAACL-HLT, 2010. pdf
- “Scalable Inference and Training of Context-Rich Syntactic Translation Models,” (M. Galley, J. Graehl, K. Knight, D. Marcu, S. DeNeefe, W. Wang, and I. Thayer), Proc. ACL-COLING, 2006. pdf
- “An Overview of Probabilistic Tree Transducers for Natural Language Processing,” (K. Knight and J. Graehl), Proc. CILing, Lecture Notes in Computer Science, Springer, 2005. doi
- “Training Tree Transducers,” (J. Graehl and K. Knight), Proc. HLT-NAACL, 2004. pdf
- “Machine Transliteration,” (K. Knight and J. Graehl), Proc. ACL-EACL, 1997. pdf — *ACL Test-of-Time Paper Award, 2022*

TUTORIALS, DEMOS, REPORTS, AND PREPRINTS

- “Machine Translation,” (K. Knight and J. Graehl), Tutorial, ANLP/NAACL, 2000.

- “hyp: A Toolkit for Representing, Manipulating, and Optimizing Hypergraphs,” (M. Dreyer and J. Graehl), Proc. NAACL-HLT: Demonstrations, 2015. pdf
- “Context-free algorithms,” (J. Graehl), arXiv:1502.02328, 2015. pdf
- “Models of Synchronous Grammar Induction for SMT,” (P. Blunsom, T. Cohn, A. Lopez, C. Dyer, J. Graehl, J. Botha, V. Eidelman, et al.), Final report, JHU CLSP Summer Workshop (WS’10), 2010. workshop

Patents

- Training Tree Transducers for Probabilistic Operations, US 7,698,125 B2, Apr. 13, 2010 (J. Graehl and K. Knight)

Software

- **Carmel** finite-state transducer toolkit — github.com/graehl/carmel
- **hyp** C++ toolkit for weighted hypergraphs (with M. Dreyer) — github.com/sdl-research/hyp
- **agents** — for coding, research, and writing agents: compact routed boot, run and session tooling, agent-facing CLI conventions, skills — github.com/graehl/agents
- **Yep Anywhere** (co-author) — self-hosted, mobile-first web UI for agent sessions — github.com/kzahel/YepAnywhere

Education

University of Southern California

B.S., COMPUTER ENGINEERING AND COMPUTER SCIENCE

1994–1998